

Fairness, Bias, and Ethics in ML

As machine learning systems affect billions of people, it is critical to ensure they are fair, free from bias, and developed ethically.

1. Unacceptable Historical Failures

There are multiple well-documented cases of ML systems exhibiting unacceptable bias, often causing real-world harm:

- **Hiring:** A hiring tool was found to **discriminate against women**.
- **Face Recognition:** Systems were shown to incorrectly match **dark-skinned individuals** to criminal mug shots far more often than light-skinned individuals.
- **Loan Applications:** Bank loan approval systems were found to be biased and **discriminate against certain subgroups**.
- **Stereotyping:** Models can **reinforce negative stereotypes** (e.g., in image search results for certain professions).

2. Adverse Use Cases

Beyond bias, ML models can be used for explicitly harmful or unethical applications:

- **Deepfakes:** Generating fake videos or audio (e.g., of political figures) without consent or disclosure.
 - **Spreading Toxic Content:** Social media algorithms, when optimizing purely for user engagement, can end up amplifying toxic or incendiary speech.
 - **Bots:** Generating fake content for commercial (fake product reviews) or political purposes.
 - **Fraud & Spam:** ML is used by bad actors to commit fraud or create more effective spam.
-

3. A Practical Framework for Ethical ML

Ethics is a complex field that humanity has studied for thousands of years; there is **no simple checklist** that guarantees a system is ethical. However, the following practical steps can provide guidance and help identify and mitigate harm.

1. Assemble a Diverse Team

- A diverse team (across gender, ethnicity, culture, etc.) is better at **brainstorming potential harms**, especially to vulnerable groups.
- This diversity increases the odds that potential problems will be spotted *before* the system is deployed.

2. Conduct a Literature Search

- Look for **existing standards or guidelines** relevant to your specific industry or application (e.g., emerging standards in finance for fair loan applications).

3. Audit *Before* Deployment

- This is a crucial line of defense.
- After training the model but *before* deploying it, **audit the system** for the potential harms you brainstormed.
- *Example:* Measure the model's error rate and performance *separately* for different genders, ethnicities, or other identified subgroups to check for bias.

4. Develop a Mitigation Plan

- Have a plan in place **before a problem occurs**.
- *Example:* Self-driving car teams have plans for what to do if a car is in an accident.
- A simple plan could be to **roll back** to a previous, safer version of the system if harm is detected.

5. Monitor *After* Deployment

- Continuously monitor the system *after* it is live.
- This allows you to act quickly and trigger your mitigation plan if a new problem (like bias or poor performance on new data) is detected.

Context is Key

Not all projects have the same level of ethical risk.

- **Low Stakes:** A model to optimize coffee bean roasting has low ethical implications.
- **High Stakes:** A model that approves bank loans or assists in medical diagnoses has very high ethical implications, and these steps are critical.